Supplementary Material for VISTA: Situated Visual Modality Adaptation for Spoken VLM Responses in XR

Category: Research

S1 Additional Implementation Details

Table S1 summarizes the run-time models, services, and client components that complement the pipeline description in the main paper. The prompt instructions and output schemas for segmentation, modality assignment, and cue generation are provided in Section S2.1.

Table S1: Stage-level implementation details corresponding to the system walkthrough in the main paper.

Stage	Implementation
❶ Input	Samsung Galaxy XR; Camera2 egocentric capture at 1 fps; OpenXR hand input; server-side voice activity detection.
❷ Response Generator	Gemini 3 Flash conditioned on the speech query and latest egocentric frame.
❸ Ack Generator	Gemini 2.5 Flash Native Audio for immediate spoken acknowledgment.
❹ Response Segmenter	Prompted VLM with Pydantic-validated structured output for segments, keywords, response–scene-relation labels, and overview metadata.
❺ Object Detector	VLM-based target detection returning normalized bounding boxes.
❻ Gist TTS	Google Cloud TTS for overview narration.
❼ Overview Delivery	Unity `Physics.Raycast` for world-space anchoring and a client-side Sync Manager for coordinated UI and gist playback.
❽ Modality Assigner	Prompted VLM with Pydantic-validated cue assignments, execution specifications, and spoken narration.
❾ Visual Modality Generator	VLM-based localization and arrow planning; SerpApi and Gemini 3.1 Flash for reference imagery; Meshy6 and GLTFast for 3D assets.
❿ Audio Modality Generator	SSML timing-mark generation and Google Cloud TTS with millisecond-level mark timestamps.
⓫ Segment Playback	Unity-based Sync Manager that activates each visual element using its corresponding TTS timestamp.

S2 System Prompts and Schemas

All stages use Pydantic[1]-validated JSON schemas. Below, we show the core instructions for each stage. Prompts and schemas are reproduced verbatim from the implementation; the schema field `referent_availability` and the prompt term "referent state" correspond to the response–scene relation label used in the paper.

S2.1 Response Segmenter ❹

```
You are given: (1) the user's query, (2) the full response
text, and (3) the user's egocentric images.
Decompose the response into N semantic segments. Place a
boundary when the dominant referent changes or when it
shifts between something the user can see and something
they cannot. Do not mirror the response's own section
structure; one original section may need splitting if it
mixes visible and non-visible referents.
Per segment, output:
• keyword: Extract a concise 1--3 word topic label that
  summarizes this segment.
• content: Copy the verbatim text span from the original
  response.
• availability_reasoning: Think step-by-step:
  1. Identify the main target of this segment. What
     entity, region, or concept does it primarily talk
     about?
  2. Is the main target conceptual, numerical, or abstract
     --- something with no physical form that could be
     pointed at? If yes → abstract. Stop here.
  3. Look at the user's egocentric images. Can you see
     the main target? If it is not visible → absent.
     Stop here.
  4. The target is visible. Would highlighting it help
     the user locate or distinguish it in the scene? If
     yes → visible.
• referent_availability: The label you arrived at above.
Global output:
• segment_count: How many segments you produced.
• main_target: The single most prominent object or region
  that the entire response is about. This will be used to
  anchor the overview UI in the user's view.
• gist_text: Write a one-sentence spoken summary that
  naturally includes the segment count (e.g., ''This is
  The Last Supper, and I can explain it through 5 main
  topics.'').
• core_keyword: Extract a 2--3 word label for the entire
  response.
```

S2.2 Modality Assigner ❽

```
You are given: (1) the segment content, (2) its
referent_availability label from Response Segmenter, and
(3) the user's egocentric images.
Read the segment and list every element that deserves its
own visual cue. All elements in a segment share the same
referent state. For each element, reason through the
following:
Per-element reasoning:
1. Modality. The referent state determines the base
   modality: visible → spatial_highlight (the user can see
   it; highlight where it is), absent → reference_image
   (the user cannot see it; show what it looks like),
   abstract → keyword (there is nothing to point at; pin
   the term on screen). For present and absent segments,
   also identify terms, values, or proper nouns that the
   listener would struggle to retain from audio alone and
   assign them as additional keyword elements.
2. Extended modality. Does this element describe a motion
   or directional relationship between two targets? → add
   3D_arrow. Does it reference a place the user cannot
   physically visit but would benefit from experiencing
   spatially? → add panoramic_view. Does the user need to
   compare the referent's physical scale or shape against a
   real counterpart? → add 3D_object.
```

[1] https://docs.pydantic.dev/

3. *Execution spec.* Describe the target in enough detail for a downstream handler to act on. Be specific: ‘‘the fourth person from the left at the table’’ for a spatial highlight, ‘‘the refectory of Santa Maria delle Grazie with The Last Supper visible on the far wall’’ for a reference image, ‘‘date’’ or ‘‘artist name’’ for a keyword.

Per-segment output:
- base_modality: spatial_highlight | reference_image | keyword.
- elements: Collect all elements into a flat, ordered list with sequential IDs: visual_0, visual_1, visual_2, ... Keywords are listed as their own elements alongside spatial highlights or reference images, each with its own ID. For each element, include: its type (spatial_highlight, reference_image, or keyword), its execution spec, and any extended modality flags. For Abstract segments, all elements are keywords.
- narration: Rewrite the segment text for spoken delivery. Replace redundant object names with deictic references that pair with the visual cues (e.g., ‘‘this figure right here,’’ ‘‘as you can see’’).

S2.3 Visual Modality Generator ❾

The Visual Modality Generator invokes the following handlers to generate the visual cues selected by the Modality Assigner.

S2.3.1 Spatial Highlight Handler

You are given: (1) one or more object descriptions from the execution spec and (2) the user’s egocentric image. Detect the specified objects in the image and return a bounding box for each. If multiple object names are provided (comma-separated), detect each one independently.

Rules:
1. Return exactly one bounding box per requested object. If multiple instances of the same object are visible, select the clearest one.
2. Only detect objects that exactly match the requested names. Do not hallucinate or detect similar but unrequested objects.
3. If a requested object is not visible in the image, omit it from the output.

Per-object output:
- box_2d: [ymin, xmin, ymax, xmax] normalized to $[0, 1000]$.
- label: the exact object name from the request.
- visibility_quality: a float in $[0, 1]$ rating how well the object is visible.
  - 1.0 = fully visible, clear, well-lit, unoccluded.
  - 0.7–0.9 = mostly visible with minor occlusion or lighting issues.
  - 0.4–0.6 = partially occluded or unclear.
  - 0.0–0.3 = heavily occluded, cut off at edges, or barely visible.
- quality_reasoning: brief justification for the quality score (e.g., ‘‘fully visible,’’ ‘‘hand blocking left side,’’ ‘‘cut off at top edge’’).

S2.3.2 3D Arrow Handler

You are given: (1) the execution spec from Modality Assigner and (2) the user’s egocentric image.
The client has a pre-defined library of arrow primitives. Your job is to select the right combination so the arrow conveys the described motion or direction.

Output the following for each arrow:
1. topology: Where does the arrow start and end? target-to-target (both endpoints are objects in the image), target-to-midair (starts at an object, ends in empty space), or midair-to-target (starts in empty space, ends at an object).
2. path_style: What shape should the path take? Choose from straight, arc, loop, spiral, zigzag.
3. flow_speed: How fast should the flow animation move? slow, normal, or fast.
4. start_anchor, end_anchor: Locate the start and end points in the image as normalized $[x, y]$ coordinates.
5. midair_params (only if one endpoint is mid-air): Specify the direction the arrow extends into (e.g., down, upright, forward) and the length in meters (e.g., 0.1m).

S2.4 Audio Modality Generator ❿

You are given: (1) the narration text and (2) the ordered list of visual elements with their IDs from Modality Assigner.
Insert timing marks into the narration so the client knows exactly when to trigger each visual cue during audio playback.

Rules:
1. Find where each element is first mentioned in the narration. Insert <mark name="<id>"/> (e.g., visual_0, visual_1) immediately before that mention.
2. For 3D arrows, insert <mark name="arrow_start"/> before the action clause where the arrow animation should begin.
3. Insert <break> after each mark, scaled to the element’s type: 300ms for spatial highlights (the user’s gaze needs to saccade to the new target), 750ms for reference images (the user encounters unfamiliar content for the first time), 300ms for arrow lead-ins (the arrow appears just before the clause), and no pause for keywords (they echo what is spoken).
4. The number of marks must exactly match the number of elements. Use sequential IDs and do not skip any.

Example (Present, three spatial highlights):

```
<speak><mark name="visual_0"/>Jesus forms a
stable triangular shape,<break time="300ms"/>
while the apostles are arranged in
<mark name="visual_1"/>the first,
<break time="300ms"/>
and <mark name="visual_2"/>third groups.</speak>
```

S3 Pipeline Evaluation Scenarios and Judgment Criteria

To identify everyday scenarios in which users would want rich, multimodal responses from an XR agent, we conducted a formative workshop ($N = 6$). XR glasses are not yet widely adopted, so we started from participants’ existing VLM usage to identify when people request information-dense responses about their real-world surroundings. We built a local web viewer that runs on each participant’s own device to keep their data private (Figure S1a–d). Before attending, participants exported their conversation histories from ChatGPT, Gemini, or both. They loaded these into the web viewer, browsed their past queries, and selected cases where they had asked about something physically in front of them and received a detailed, multi-part response. From these cases, six recurring categories emerged: Art & Culture, Food & Cooking, Shopping, Home Repair, Home & Personal, and Learning & Interpretation. For each category, we authored four to six scenarios grounded in specific real-world settings, yielding 30 evaluation scenarios in total.

S3.1 Judgment Criteria

Two authors independently judged each pipeline output as correct or incorrect against the criteria below, and a third author adjudi-

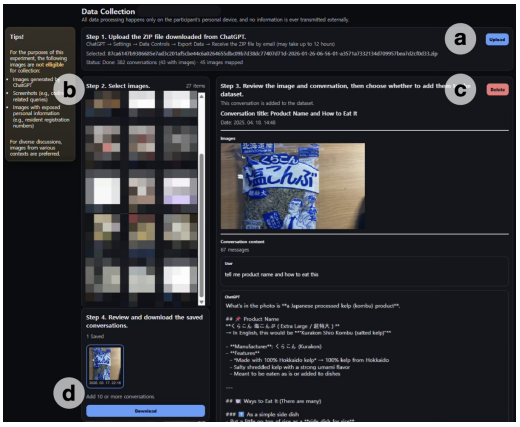


Figure S1: A locally hosted web viewer that lets participants collect their personal VLM conversation data. (a) Importing conversation histories from everyday VLM services. (b) Browsing cases with image-based queries. (c) Inspecting individual conversations. (d) Selecting and exporting cases for the workshop.

Table S2: The 30 scenarios used for the pipeline evaluation, grouped by category. Each row shows the scene description followed by the user query. The list continues in Table S3.

#	Scene / Query
	Art & Culture
1	Single Western painting on the wall. → Tell me about this painting.
2	East Asian calligraphy or ink painting. → Can you interpret this for me?
3	Sculpture or installation art. → What is this?
4	Historic building exterior. → What's the history behind this building?

cated the items on which they disagreed. The criteria instantiate the response–scene-relation definitions in Section 3 of the main paper.

- **Response Segmenter.** A segment is correct when its boundaries follow response–scene-relation transitions or breaks in semantic coherence, and its label matches the actual relation between the segment's main target and the egocentric scene. A conditional or future state that cannot be seen in the current view is Absent, not Visible, even when it results from acting on a visible object.
- **Modality Assigner.** The assigned base and extended visual modalities must match the mapping in Section 3.2 of the main paper for the segment's response–scene relation and content (e.g., a described motion warrants a 3D arrow, and a described place warrants a panoramic view).
- **Spatial highlight.** The highlight must cover the described physical target(s). Partial coverage of a described set or a highlight on a different object counts as incorrect.
- **3D arrow.** The arrow must depict a meaningful, scene-grounded motion or direction, with start and end points aligned to the described trajectory. An arrow that terminates in midair or contradicts its paired highlight counts as incorrect.
- **Reference image, panoramic view, and 3D object.** The visual must depict the specified missing entity or target state in the described context. A generic image of the right category that misses the described aspect counts as incorrect.

Table S3: Continuation of Table S2: scenarios 5–30.

#	Scene / Query
	Food & Cooking
5	Pour-over coffee setup on counter. → How do I brew pour-over with this?
6	Vegetables and seasonings around cutting board. → What can I make with these?
7	Side dishes on the table. → Tell me about these side dishes.
8	Baking ingredients on table. → I want to make macarons. How do I start?
9	Spice jars and bottles on shelf. → What are these spices, and how should I use them?
10	Fridge with food. → What can I make for dinner with what's in here?
	Shopping
11	Convenience store beverage shelf. → Can you recommend a zero-calorie drink?
12	Ramyeon display. → Which ones aren't spicy?
13	Wine shelf. → Which one goes well with steak?
14	Street with restaurant signs. → Any good restaurants around here?
	Home Repair
15	Shelf parts and hardware on floor. → How do I put this shelf together?
16	Candle jar with candle care tools. → How do I use these candle tools?
17	Sewing machine with fabric. → How do I make a pouch with this?
18	Bicycle flipped upside down. → How do I change this tire?
19	Printer showing error on display. → How do I fix this?
	Home & Personal
20	Balcony with multiple potted plants. → How do I take care of these plants?
21	Multiple medications on table. → How should I take these?
22	Photo of items to pack. → I have too much stuff. What should I leave out?
23	Home gym equipment in living room. → Can you put together a workout routine with these?
24	Car dashboard with warning lights. → What do these warning lights mean?
25	Living room with mixed objects. → Is there anything dangerous here for a toddler?
	Learning & Interpretation
26	Tide pool close-up. → What are these creatures?
27	Textbook diagram (circuit or cell). → Can you explain this diagram?
28	Map of hiking trail. → Which trail is best for beginners?
29	Board game set up mid-play. → How do you play this?
30	Presentation slide on screen. → Can you explain this slide in more detail?

- **Keywords.** Keywords must preserve the actual unfamiliar term, proper name, number, or value needed to follow the narration, in the user's language. Omitting a retention-critical term or substituting a less informative one counts as incorrect.
- **Audio Modality Generator (SSML).** Timing marks must occur at the first relevant mention and correspond one-to-one with the displayed cues, in the correct order. Sequences over multiple vis-

Table S4: Example quiz generated for the Beverage Crafting task (pour-over coffee preparation, Set A). The quiz was generated from P04's session in which the participant asked, "Can you explain how to make a cup of coffee?".

#	Question	Ans.
1	What is the purpose of transferring boiling water into the stainless steel drip pot?	To help control the pour
2	How many grams of coffee beans are recommended for this brewing method?	15–20 grams
3	What consistency should the coffee grounds have after using the hand grinder?	Medium-coarse like sea salt
4	What should be done after placing the dripper and carafe on the digital scale?	Tare it to zero
5	How much water should be used for the initial "bloom" step?	40 ml
6	How long should you wait for the coffee to bloom before the final pour?	30 seconds
7	What is the total target volume of water for the entire brewing process?	250–300 ml
8	In what motion should the water be poured over the coffee grounds?	Steady, concentric circles
9	What type of dripper is mentioned in the instructions?	Mesh dripper
10	What is the primary purpose of the "bloom" phase in coffee brewing?	To release gases

ible targets need sufficient transitions, and an Absent reference cue requires the specified viewing pause.

S4 Comprehension Quiz for User Evaluation

This section presents the comprehension quiz used to objectively assess participants' retention of information after each task. We describe the prompt used to generate quizzes and provide an example quiz from the Beverage Crafting task.

S4.1 Quiz Generation Prompt

After each task, the system generated a 10-item comprehension quiz from the agent's response. We fed the user's original query and the full response text into Gemini 3 Flash[2] with the following instructions.

```
You are a quiz generator for a user study evaluating an XR
assistant system. Given the user's original question and
the assistant's response, generate exactly 10
multiple-choice questions to test comprehension of the
response.

Each question must have exactly 4 choices (A--D) with one
correct answer. Questions should vary in difficulty, cover
different parts of the response, and use plausible
distractors. Keep each question under 150 characters.

Input: User's question and the assistant's full response
text.
Output: JSON array of 10 {question, choices,
correct_answer} objects.
```

S4.2 Example Quiz

Table S4 shows the quiz generated from one participant's session in the Beverage Crafting task.

S5 Full Statistical Results of the User Study

Table S5 reports the full test statistics for the comprehension quiz and all questionnaire measures in the user study.

[2] https://deepmind.google/models/gemini/flash/

Table S5: Full statistical results of the user study. Values are participant-level means across the two tasks per condition ($N = 12$). W is the Wilcoxon signed-rank statistic, p_{Holm} the Holm-adjusted p-value, and r the matched-pairs rank-biserial correlation. The two primary outcomes were corrected as one family, and exploratory measures were corrected within each scale family.

Measure	VISTA M (SD)	Baseline M (SD)	W	p	p_{Holm}	r
Primary outcomes						
Comprehension quiz accuracy	6.92 (1.93)	5.75 (2.01)	11.0	0.048	0.048	+0.67
NASA-TLX overall workload	3.14 (0.83)	4.04 (0.79)	9.0	0.016	0.032	−0.77
Perceived knowledge gain						
Gained new information	6.25 (0.78)	4.92 (0.85)	0.0	$< .001$	0.002	+1.00
Improved understanding	5.96 (0.96)	4.67 (1.07)	1.0	0.004	0.008	+0.96
Thought more deeply	5.17 (1.11)	3.79 (1.08)	4.0	0.008	0.008	+0.88
Perceived information quality						
Relevance	6.04 (0.54)	5.62 (1.00)	8.5	0.125	0.125	+0.62
Detail	6.42 (0.67)	5.67 (0.83)	2.5	0.010	0.020	+0.91
Clarity	5.62 (1.17)	3.75 (1.47)	1.5	0.003	0.009	+0.95
NASA-TLX subscales						
Mental demand	3.08 (1.43)	4.96 (1.32)	0.0	$< .001$	0.006	−1.00
Physical demand	2.75 (1.66)	3.12 (1.69)	13.0	0.508	0.508	−0.28
Temporal demand	2.50 (1.55)	4.58 (1.33)	4.0	0.007	0.021	−0.88
Performance	5.25 (1.20)	3.21 (1.36)	0.0	$< .001$	0.006	+1.00
Effort	3.00 (1.13)	4.33 (1.09)	4.5	0.031	0.062	−0.80
Frustration	2.25 (1.14)	4.04 (1.36)	5.0	0.005	0.020	−0.87